

Let's learn about **satellites!**

Space mission and **satellite** design course

Satellite AIT part 1

AIT – Assembly integration and testing

Assembly of many components, parts and manufactured subsystems

Integration of the subsystems to build the spacecraft

Test of the flight spacecraft

Why AIV – Assembly Integration and Verification?

Objective: Verify to a very high level of **confidence** and **probability** that the hardware will perform the desired mission

AIT – Assembly integration and testing

Steps:

- ✓ AIV concepts and terminology
- ✓ Verification VS Validation
- ✓ Verification methods
- ✓ Verification matrix
- ✓ FLATSAT EM PFM FM
- ✓ Integration environment
- ✓ Assembly and integration process

AIT – Definitions

Assembly – the process of **mechanically bringing together** hardware components large and small, from many suppliers to build up the deliverable spacecraft.

Hardware components: s/c structures, propulsion modules, electronic board, payload sensors, cables, thermal blankets, bolts and washers

Processes and specifications: for each phase of integration, and for **special processes** (adhesive, bolt torquing, special coating)

AIT – Definitions

Integration – the process of **physically and functionally combining component equipments**, in a **controlled** and **measured** sequence. □ making sure it all work together.

For example: power go to the right place? Output V of A is input V of B? No obstruction for optical payload or for deployables. Software/hardware interface

Important note: when components come from **different contractors** errors occur □ interface discrepancies

AIT – Definitions

Verification – the total process by which **conformance to all applicable requirements** are demonstrated.

Verification = Qualification + Acceptance

Qualification – demonstrating that the **spacecraft design** is fully capable of meeting all the applicable requirements with **margins**

- Qual. tests include «over test»
- Qualification tests not on the flight hardware
- Qualification can be transferable

AIT – Definitions

Acceptance – demonstration that the end product, the flight hardware, is **free from defects**.

Defects:

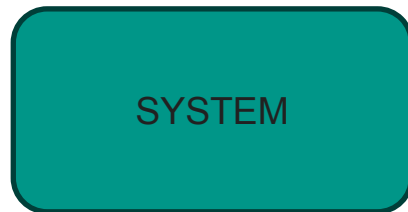
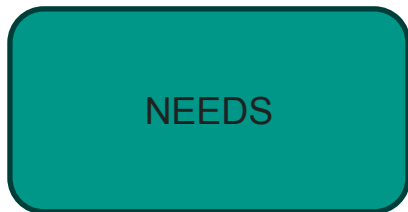
- Workmanship defects
- Material defects
- assembly defects

Confirm that the **flight hardware** has been built according to the **qualified** design

AIT – Definitions

Validation is a process of confirming that a **set of requirements**, design or system meets the **intent** of the developer or customer.

Verification is a process of confirming that **a requirement** or system is **compliant**.

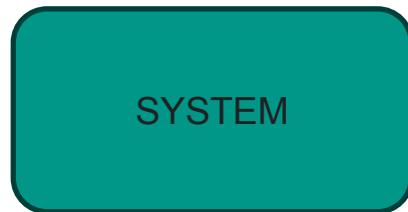
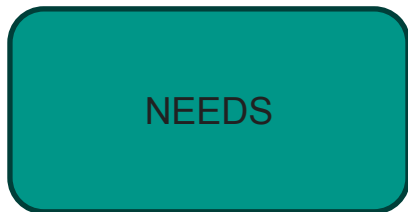


Design is **validated** against the needs and the system is **verified** against the design

AIT – Definitions

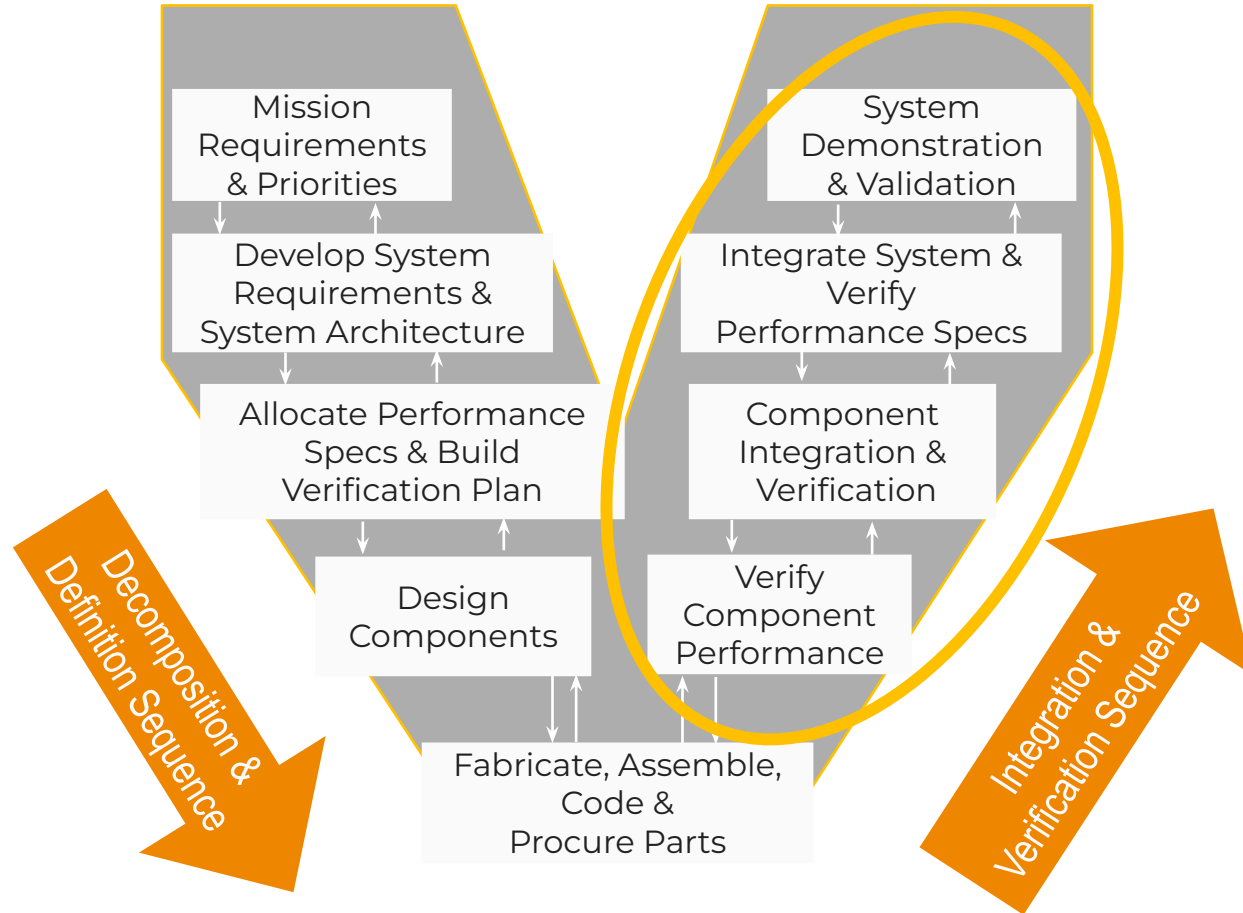
Validating a system - Building the **right system**: Ensuring that the system will satisfy the actual **needs** of the **stakeholders**.

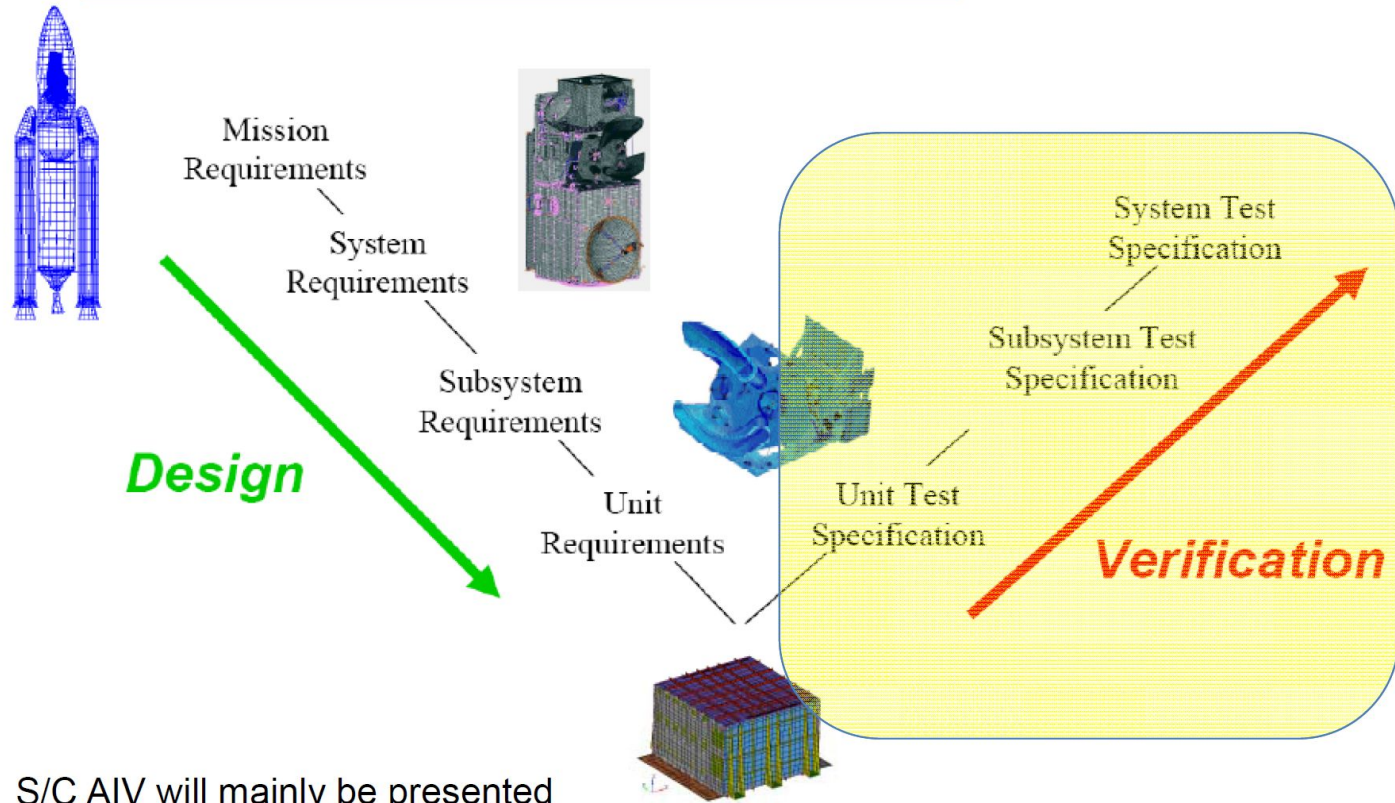
Verifying a system- Building the **system right**: ensuring that the system complies with the system requirements and conforms to its design.



Design is **validated** against the needs and the system is **verified** against the design

AIT – Definitions





AIT – Definitions

Verification methods

There are 4 methods for verifying a requirement:

- 1) Inspection
- 2) Analysis
- 3) Review of design
- 4) Test

What is the most important one?

AIT – Definitions

Verification methods – Inspection

Inspections determine conformance to requirements by the **visual examination** of drawings, data, or the item itself using **standard quality control methods**, without the use of special laboratory procedures or equipment.

□ physical attribute such as dimensions, weight, physical characteristics, color or markings, etc.

Examples:

“...shall be 24 mm long $\pm 0,1$ mm ...”

“...shall be painted white...”

“...the roughness shall be Ra 0,8”

AIT – Definitions

Verification methods – Analysis

Analysis is the evaluation of data by **analytical techniques** to determine that the item will meet specified requirements.

Analysis techniques: systems engineering analysis, statistics, modeling, similarity and simulation.

Analysis is used when the system **cannot be tested** or It is **cost-effective**

AIT – Definitions

Verification methods – Test

Test consists of operation of **all** or **part** of the system under a limited set of controlled conditions to determine that **quantitative design** or **performance requirements** have been met

Use of **elaborate instrumentation** and **special test equipment** to measure the parameter

Tests can be performed at **any level** of assembly within the system assembly hierarchy.

The kind of language used in the item **requirement** that usually indicates verification by test is:

“...shall provide 50 Hz...”

“...shall be settable over a range of 0 to 30 degrees C...”

AIT – Definitions

the top three verification methods are test, test, and test

one test is worth a thousand expert opinions

Verification methods - summary

- 1) **Test:** preferred method; simulation of an outside environment (vibration, thermo, vacuum, radiation) and verification of the correct operation of the satellite (switch-on, send telemetry, open deployable solar panels)
- 2) **Analysis:** mathematical modelling and computer simulation. Only when test is not possible or too expensive
- 3) **Review of design:** looks at approved design reports. I.e. a specific design is already qualified and no further qualification analysis or testing is required
- 4) **Inspection:** determine conformance to specific features. I.e. measure with a gauge a quote, check the position of a hole, check the materials used. Inspection is performed by Quality Control department

Example
Shaker, Acoustic chamber, etc.
Esatan, Nastran, STK, etc.
With official reports the design is verified, etc.
Visual verification

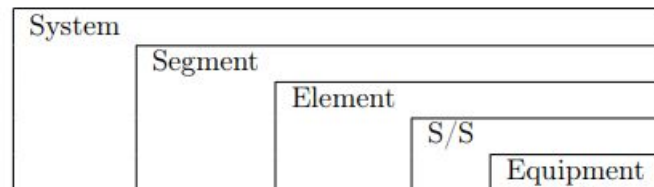
AIT – Definitions

Verification levels

Spacecraft level

Module or subsystem level

Component level



Important: Verification at **spacecraft level is the best** but...

□ The **sooner** you detect anomalies, the **cheaper** it is

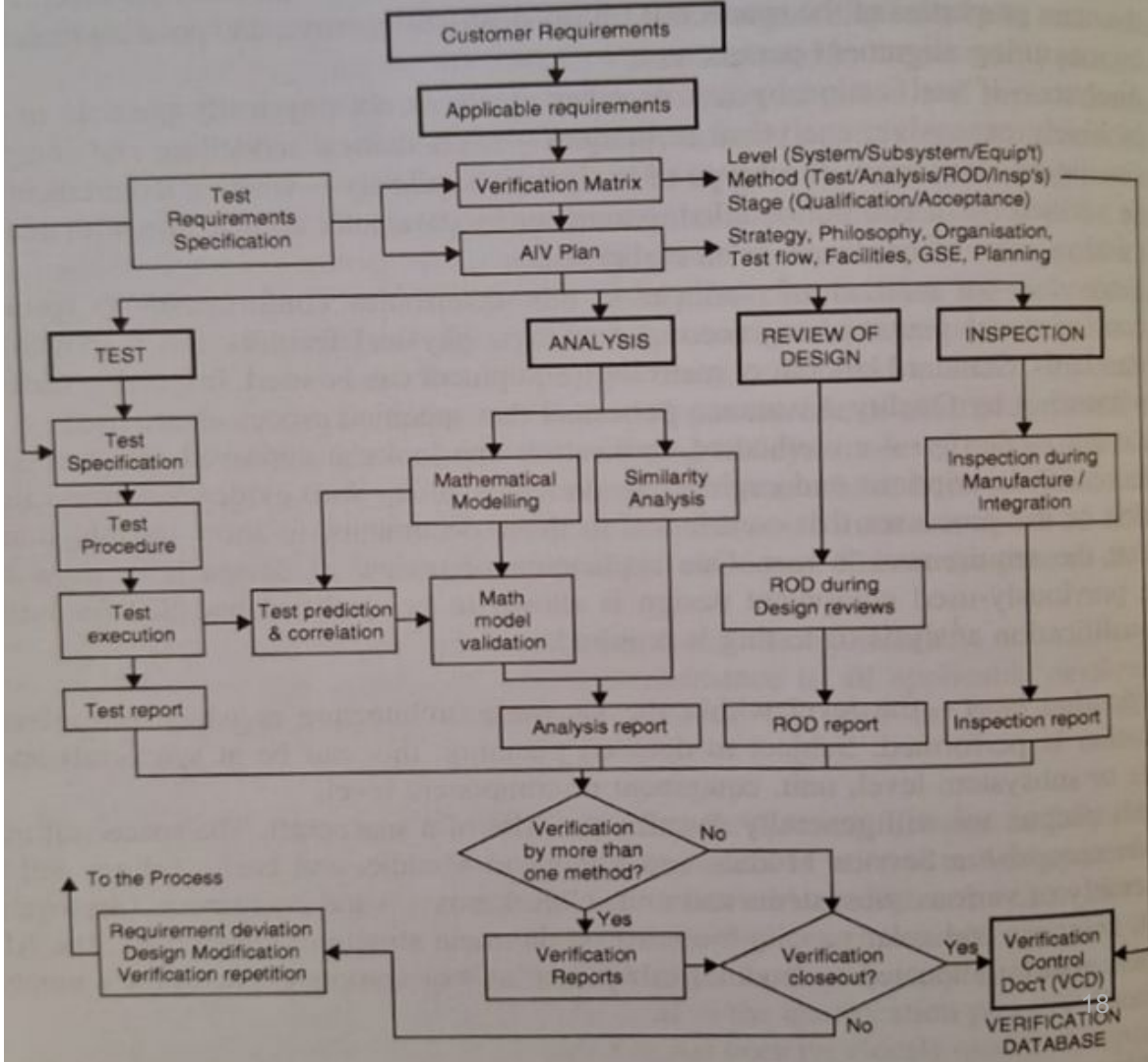
□ Testing of **smaller** items is **easier** and **cheaper**

...evaluate to re-verify the performance throughout the programme

AIT

Verification plan

ECSS-E-ST-10-02C



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Verification matrix

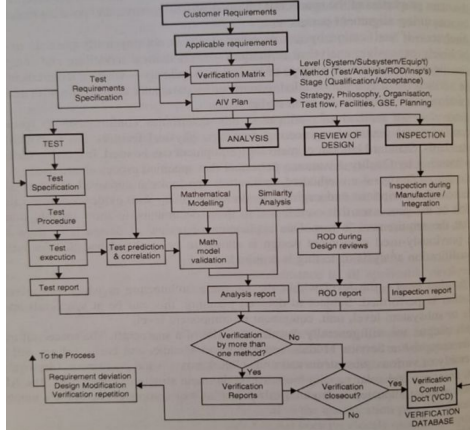


TABLE D-1 Requirements Verification Matrix

Requirement No.	Document	Paragraph	Shall Statement	Verification Success Criteria	Verification Method	Facility or Lab	Phase ^a	Acceptance Requirement?	Preflight Acceptance?	Performing Organization	Results
<i>Unique identifier or each requirement</i>	<i>Document number the requirement is contained within</i>	<i>Paragraph number of the requirement</i>	<i>Text (within reason) of the requirement, i.e., the "shall"</i>	<i>Success criteria for the requirement</i>	<i>Verification method for the requirement (analysis, inspection, demonstration, test)</i>	<i>Facility or laboratory used to perform the verification and validation.</i>	<i>Phase in which the verification and validation will be performed.</i>	<i>Indicate whether this requirement is also verified during initial acceptance testing of each unit.</i>	<i>Indicate whether this requirement is also verified during any pre-flight or recurring acceptance testing of each unit</i>	<i>Organization responsible for performing the verification</i>	<i>Indicate documents that contain the objective evidence that requirement was satisfied</i>
P-1	xxx	3.2.1.1 Capability: Support Uplinked Data (LDR)	System X shall provide a max. ground-to-station uplink of...	1. System X locks to forward link at the min and max data rate tolerances 2. System X locks to the forward link at the min and max operating frequency tolerances	Test	xxx	5	Yes	No	xxx	TPS xxxx
P-i	xxx	Other paragraphs	Other "shalls" in PTRS	Other criteria	xxx	xxx	xxx	Yes/No	Yes/No	xxx	Memo-xxx

AIT Verification matrix

EXAMPLE - The verification matrix specifies:

Requirement: “Space vehicle first-mode natural frequency shall be greater than 35 Hz.”

Associated verification requirement, including success criteria: “The space vehicle first-mode natural frequency shall be verified by test. The test shall conduct a modal survey of the vehicle using a vibration table. The test shall be considered *successful* if the measured first-mode is greater than 35 Hz.”

Method of verification: test

Level verification is to be performed: spacecraft

Who performs the verification

AIT – FLATSAT EM PFM FM



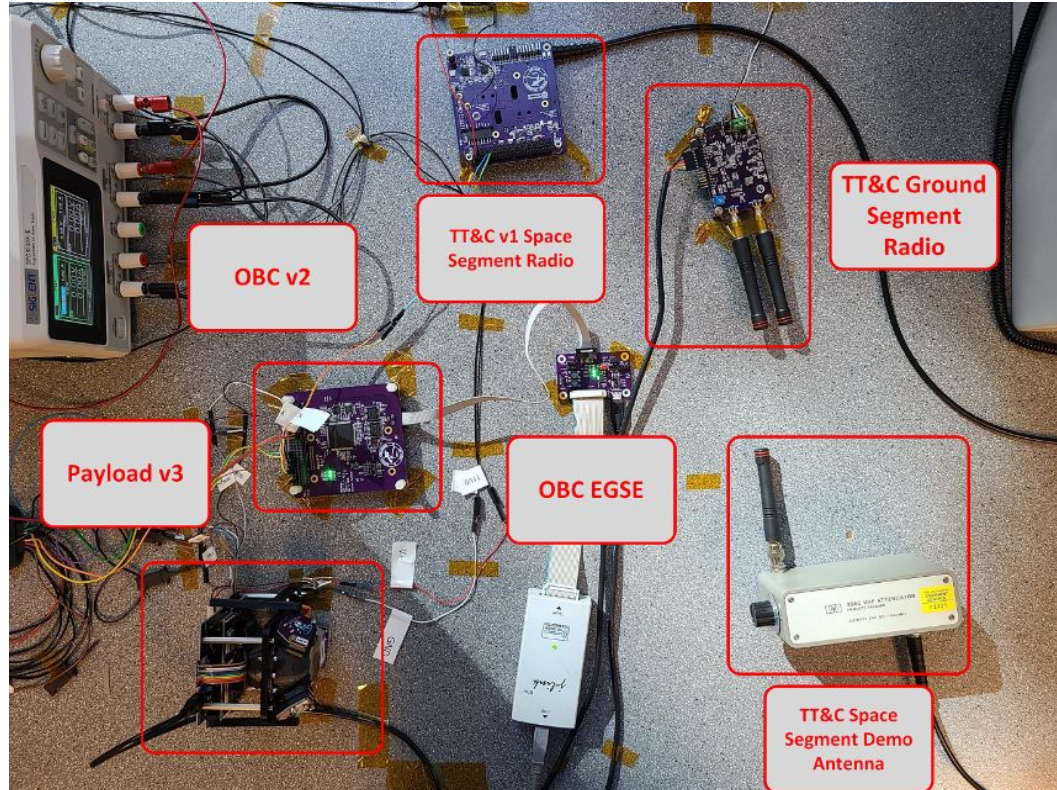
Flatsat

- disassembled CubeSat
- Easy to access
- Easy to perform tests and verify functionality of each subsystem
- Change subsystems in a quick way
- Modularity

In the picture you can see:
on-board computer,
radio and telemetry,
attitude and orbit control system,
battery and power system and
satellite navigation receiver,
with additional slots to plug in
external hardware

Example of a Flatsat (image credit: ESA)

AIT – FLATSAT EM PFM FM



(image credit: ORCASAT)

- Simulate and verify all the satellite functions (downlink, data command, payload functions)

In the picture there isn't an electrical power system and it is used a generator □ it is possible to build the flatsat and start testing in advance, and after receiving parts from suppliers it is possible to add them to the Flatsat

AIT – FLATSAT EM PFM FM

Link video:

<https://www.youtube.com/watch?v=z7ZrC2Zlbw&t=187s>



(video credit: ORCASAT)

AIT – FLATSAT EM PFM FM



Engineering model of payload GALILEO (image credit: ESA)

Definition from ECSS-S-ST-00-01C

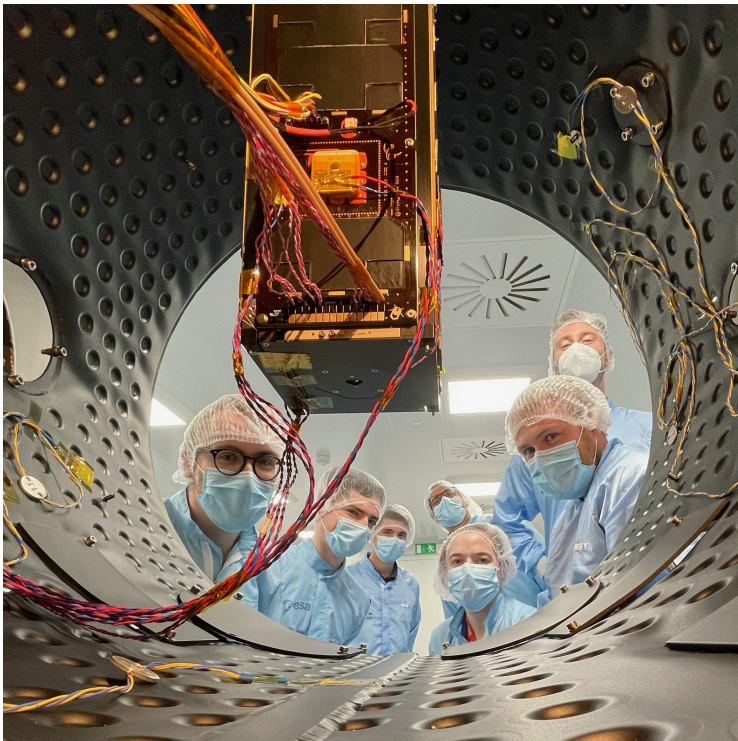
Engineering Model (EM):

flight representative model in terms of form, fit and function used for functional and failure effect verification

The engineering model is usually not equipped with high reliability parts or full redundancy. (It can use a mixture of flight-grade components (usually spare or out-of-life in stores) and commercial components).

The engineering model is also used for final validation of test facilities, GSE and associated procedures.

AIT – FLATSAT EM PFM FM



(image credit: EIRSAT-1)

Definition from ECSS-S-ST-00-01C

Qualification Model (QM):

model, which fully reflects all aspects of the flight model design, used for complete functional and environmental qualification testing

This is the model that is subjected to the qualification campaign.

A qualification model is only necessary for newly-designed hardware or when a delta qualification is performed for adaptation to the project.

The qualification model is not intended to be used for flight, since it is overtested.

AIT – FLATSAT EM PFM FM



GALILEO FM – Acoustic testing (image credit: ESA)

Definition from ECSS-S-ST-00-01C

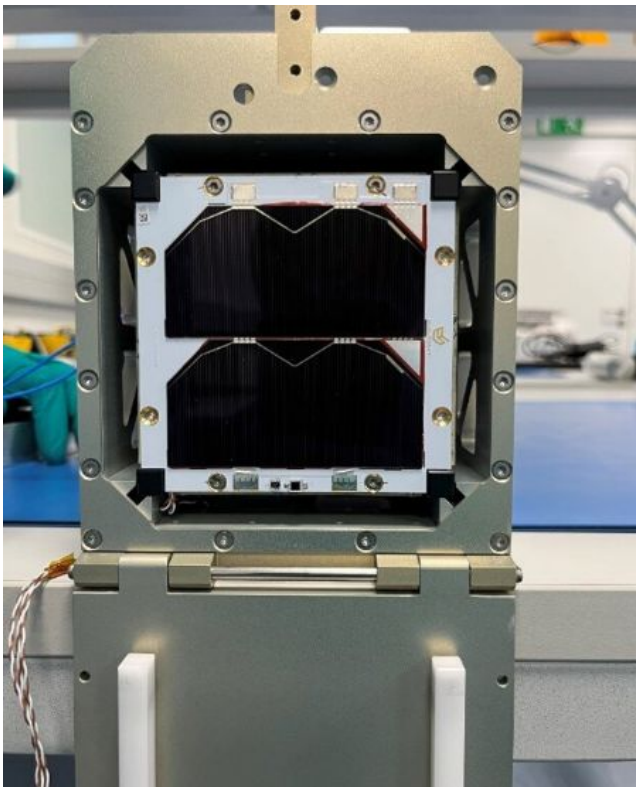
Flight Model:

end product that is intended for flight

The model actually flying; this is tested to acceptance-level testing.

The flight model is subjected to formal functional and environmental acceptance testing.

AIT – FLATSAT EM PFM FM



(image credit: ISTSat)

Definition from ECSS-S-ST-00-01C

Proto Flight Model:

flight model on which a partial or complete protoflight qualification test campaign is performed before flight

Frequently used in the New Space Economy in order to reduce time and costs.

A protoflight approach carries a higher technical risk approach compared to a full qualification test program

Test levels higher than the acceptance campaign but lower than Qualification campaign

AIT – Integration facilities (Clean Room)

Clean room

Room in which:

- the concentration of airborne particles is controlled,
- constructed and used to minimize the introduction, generation and retention of particles inside the room,
- relevant parameters, e.g. temperature, humidity and pressure, are controlled as necessary

Clean room □ ISO 14644-6

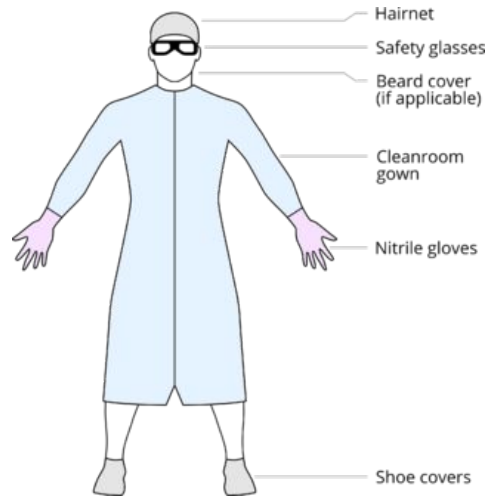
Cleanliness and contamination control □
ECSS-Q-ST-70-01C



AIT – Integration facilities (Clean Room)

Clean Room levels

Dress code



Class	Maximum Particles/ft ³					ISO equivalent
	≥0.1 µm	≥0.2 µm	≥0.3 µm	≥0.5 µm	≥5 µm	
1	35	7	3	1		ISO3
10	350	75	30	10		ISO4
100				100		ISO5
1000				1000	7	ISO6
<u>10,000</u>				10,000	70	ISO7
100,000				100,000	700	ISO8

Changing room



Assembly and integration



https://www.youtube.com/watch?v=5zbTj0-B4_Y-
CUBESAT ASSEMBLY: ORCASat Final Assembly Time-Lapse

Thanks for your attention